

# Mathematics Competition

Student Exchange Germany – China

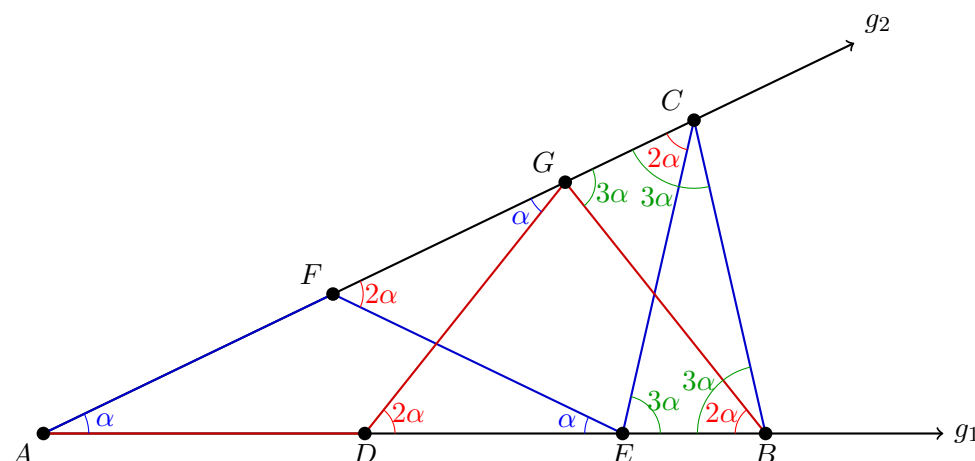
– Solution Sheet –

## Solution 1: The Seven-Segment Figure

To solve this task, we work our way through the figure from the inside to the outside. We only need two geometric rules for this:

- **Isosceles triangles:** The angles opposite the sides of equal length are equal.
- **Exterior angle theorem:** Each exterior angle of a triangle is exactly equal to the sum of the two opposite interior angles.

Since all segments ( $|\overline{AD}| = |\overline{AF}| = \dots = |\overline{CB}|$ ) have the same length according to the task, the entire zigzag figure consists solely of isosceles triangles.



The differently colored arcs in the sketch illustrate the systematic approach: For example, in triangle  $\triangle AFE$ , the **blue** angle at  $E$  is equal to  $\alpha$  because the triangle is isosceles with base  $\overline{AE}$ . The exterior angle theorem at corner  $F$  immediately gives us the next, **red** angle  $2\alpha$ . This principle repeats alternately for all inner sub-triangles, so that the angles add up exactly mathematically with each "jump" outwards and finally become the **green** angles  $3\alpha$ .

Finally, let's consider the large outer triangle  $\triangle ABC$ . It has the following three interior angles:

- Angle at  $A$ :  $\alpha$  (**blue**)
- Angle at  $B$ :  $3\alpha$  (**green** enclosing angle of triangle  $\triangle GBC$ )
- Angle at  $C$ :  $3\alpha$  (**green** enclosing angle of triangle  $\triangle EBC$ )

It is well known that the sum of angles in a triangle is always exactly  $180^\circ$ . This results in the equation:

$$\alpha + 3\alpha + 3\alpha = 180^\circ \quad \Rightarrow \quad 7\alpha = 180^\circ \quad \Rightarrow \quad \alpha = \frac{180^\circ}{7} \approx 25.71^\circ$$

**Result:** The figure closes perfectly at a starting angle of exactly  $\alpha = \frac{180^\circ}{7}$ .

## Solution 2: Large Products – Many Zeros?

### a) How many trailing zeros are formed up to 100?

A trailing zero is created by a factor of  $10 = 2 \cdot 5$ . There are more than enough twos in the product, so we only need to count the factors of 5.

The multiples of 5 are 5, 10, 15, ..., 100. These are 20 numbers, which initially provide 20 factors of 5.

The numbers 25, 50, 75, 100 each contain another factor of 5 because they are divisible by  $25 = 5^2$ . This adds four more factors of 5. In total, there are  $20 + 4 = 24$  factors of 5.

**Result:** The product  $1 \cdot 2 \cdot 3 \cdots 100$  ends with exactly **24 zeros**.

### b) When do 250 trailing zeros first appear?

Working backwards is a good strategy here. We know from part a) that the product up to 100 ends with 24 zeros. If we go up to 1000, we can expect about ten times as many trailing zeros. That probably puts us close to 250.

Therefore, we first examine the product  $1 \cdot 2 \cdot 3 \cdots 1000$  and again count the multiples of the powers of 5:

| Multiples of ... | Amount in product up to 1000 |
|------------------|------------------------------|
| 5                | 200                          |
| 25               | 40                           |
| 125              | 8                            |
| 625              | 1                            |

The multiples of 25, 125 and 625 each provide additional factors of 5. In total, you get  $200 + 40 + 8 + 1 = 249$  factors. So the product up to 1000 ends with 249 zeros.

The following numbers 1001, 1002, 1003 and 1004 do not bring a new factor of 5. It's the 1005 that is again divisible by 5, providing the still missing factor of 5 for the 250th zero.

**Result:** The 250th trailing zero appears for the first time at **n = 1005**.